Source-Based AI Recommendation Explanations — Experiment Flow v1

1. Survey Structure

The experiment will be implemented as a five-condition between-subjects online survey.

Each participant will be randomly assigned to exactly one recommendation-explanation condition:

- C1 = Control
- C2 = Behavior-Based
- C3 = Review-Based
- C4 = Product-Feature-Based
- C5 = Mixed-Source

The participant will view only one experimental stimulus.

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2. Experiment Implementation Flow

Block 1 — Informed Consent

Participants first view the informed-consent page.

Suggested elements:

- purpose described in broad terms as research on AI-assisted product recommendations;
- estimated completion time;
- voluntary participation;
- ability to discontinue participation;
- confidentiality / data-use information;
- eligibility requirements;
- researchers contact information;
- ethics/IRB information when applicable.

Consent Question

Do you agree to participate in this study?

- Yes, I agree to participate.
- No, I do not agree to participate.

Branch Logic

If **No** → End Survey.

If **Yes** → Continue.

Block 2 — Shopping Scenario

Display the common scenario to all participants.

Participants imagine that they are shopping for wireless headphones and are looking for:

- good sound quality,
- long battery life,
- comfortable fit, and
- reasonable value.

They are told that an AI shopping assistant will recommend a product.

Instruction

Please read the recommendation carefully. You will be asked several questions about the recommendation afterward.

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3. Random Assignment

Insert a **Randomizer** in Survey Flow.

Randomizer Setting

Randomly present 1 of the following 5 elements.

Use approximately even presentation across conditions.

Inside the Randomizer:

C1 — Control

Set Condition Variable:

condition = control

Then display:

Stimulus Block — Control

C2 — Behavior-Based

Set Condition Variable:

condition = behavior

Then display:

Stimulus Block — Behavior-Based

C3 — Review-Based

Set Condition Variable:

condition = review

Then display:

Stimulus Block — Review-Based

C4 — Product-Feature-Based

Set Condition Variable:

condition = product_feature

Then display:

Stimulus Block — Product-Feature-Based

C5 — Mixed-Source

Set Condition Variable:

condition = mixed

Then display:

Stimulus Block — Mixed-Source

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4. Experimental Stimulus Blocks

Each condition should use the wording finalized in:

Source-Based – Stimuli v2

All five versions should use the same:

- product;
- product attributes;
- formatting;
- font;
- interface structure;
- recommendation placement;
- visual style.

Only the explanation-source information should differ systematically across conditions.

Product

AeroSound X5 Wireless Headphones

Important Setting

After participants leave the stimulus page, they should **not be allowed to return to the previous page**.

This prevents participants from changing how they inspect the manipulation after seeing later measurement questions.

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5. Main Measurement Block

After the stimulus, participants complete the primary psychological measures.

The manipulation checks should **not** appear yet.

Section A — Perceived Relevance

Display:

- PR1
- PR2
- PR3

7-point Likert scale:

1 = Strongly disagree
to
7 = Strongly agree

Randomize the order of the three items.

Section B — Perceived Source Credibility

Display:

- SC1
- SC2
- SC3

Use the same 7-point scale.

Randomize item order within this construct.

Section C — Situational Privacy Concern

Display:

- PC1
- PC2
- PC3

Use the same 7-point scale.

Randomize item order within this construct.

Section D — Trust

Display:

- TR1
- TR2
- TR3
- TR4

Use the same 7-point scale.

Randomize item order within this construct.

Section E — Reliance Intention

Display:

- RI1
- RI2
- RI3

Use the same 7-point scale.

Randomize item order within this construct.

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6. Behavioral Reliance

After the self-report measures, ask participants to make a decision.

Question

Based on the information you reviewed, what would you most likely do?

- Choose the AeroSound X5 recommended by the AI.
- Continue searching for another pair of headphones.

Coding

behavioral_reliance = 1

if participant chooses AeroSound X5.

behavioral_reliance = 0

if participant chooses to continue searching.

This variable should be analyzed separately from Reliance Intention.

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7. Manipulation Check Block

Manipulation checks appear only after the primary outcomes and behavioral decision.

Use the finalized items from:

Source-Based – Manipulation Checks v1

Items

MC1. The recommendation was based on my previous browsing or shopping behavior.

MC2. The recommendation was based on information from other consumers' reviews.

MC3. The recommendation was based on the product's features and attributes.

MC4. The recommendation combined multiple sources of information.

MC5. The explanation clearly indicated what information source or sources were used to generate the recommendation.

Use a 7-point agreement scale.

Randomize MC1–MC4.

MC5 may remain last because it measures general source disclosure rather than recognition of a particular source category.

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8. Demographic and Control Variables

For the pilot, keep this block short.

Potential variables:

Age

What is your age?

Gender

How do you describe your gender?

Online Shopping Frequency

How often do you shop online?

- Never
- Less than once per month
- 1–3 times per month
- About once per week
- Several times per week

AI Recommendation Experience

How often have you used AI tools or AI-powered recommendation systems when making product decisions?

- Never
- Rarely
- Sometimes
- Often

- Very often

Product Familiarity

How familiar are you with wireless headphones?

1 = Not at all familiar
7 = Extremely familiar

These variables should initially be treated as descriptive or control variables rather than additional primary hypotheses.

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9. Debrief

After completing the survey, participants receive a short debrief.

The debrief should explain that the study examines whether different descriptions of the information used by an AI recommendation system influence perception of the recommendation.

Participants should also be informed that:

- AeroSound X5 is a fictional product;
- the recommendation scenario was created for research purposes;
- different participants may have received different explanation-source descriptions.

Avoid revealing this information before the participant completes the experiment.

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10. Recommended Experiment Flow

The final flow should appear approximately as follows:

Consent
↓
Common Shopping Scenario
↓
Randomizer
├── C1 Control
├── C2 Behavior-Based
├── C3 Review-Based
├── C4 Product-Feature-Based
└── C5 Mixed-Source
↓
Perceived Relevance
↓
Source Credibility

↓
Privacy Concern
↓
Trust
↓
Reliance Intention
↓
Behavioral Reliance Decision
↓
Manipulation Checks
↓
Demographics / Controls
↓
Debrief
↓
End Survey

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11. Condition Variable

At minimum, save the following variable:

condition

Possible values:

- control
- behavior
- review
- product_feature
- mixed

The experiment platform should retain a unique Response ID.

Additional derived variables such as scale averages should normally be created during data analysis rather than within the experimental platform.

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12. Pilot Quality Checks

Before launching a formal study, test whether:

1. Random assignment distributes participants correctly across all five conditions.
2. Each participant sees only one stimulus.
3. Condition labels export correctly.
4. All Likert responses export as numeric values.
5. The behavioral-reliance response exports correctly.

6. Participants cannot accidentally return to the experimental stimulus after answering later questions.
7. Manipulation-check responses differentiate the intended source conditions.
8. The full survey can be completed without confusing instructions or broken logic.
9. Survey completion time is reasonable.
10. Mobile and desktop versions both display the stimulus correctly.

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13. Current Project Status

Completed:

- Conceptual Model
- Five-Condition Experimental Design
- Stimuli v2
- Manipulation Checks v1
- Hypotheses v1
- Measurement v1
- Experiment Spec v1
- Analysis Plan v1
- Power Analysis v1
- Experiment Flow v1

Next step:

Implement the pilot on an appropriate survey or experimental platform and run an internal functionality test.